

ALEXANDER MORALES

QA Automation Engineer | Software Development & DevOps

📍 Quito - Ecuador | 📞 +593 984423820 | ✉ arielmorales.2105@gmail.com
🌐 Portfolio | 🔗 LinkedIn | 🐙 GitHub

PROFESSIONAL SUMMARY

Software Engineering student in my final semester at Escuela Politécnica Nacional, pursuing a career as a QA Automation Engineer. I believe in technology as a lever for real transformation: one that reduces operational friction and improves people's lives. My experience in QA, DevOps, and web development has taught me that quality is not a phase, but a culture. I design and automate processes ensuring software fulfills its purpose, building for scalability, reliability, and data-driven decision-making.

I have contributed to improving efficiency in critical banking systems and sparking interest in programming among youth. I thrive in projects where I can connect test automation, continuous integration, engineering best practices, and effective communication between technical and business teams, ensuring faster, more secure, and sustainable software delivery.

PROFESSIONAL EXPERIENCE

- **Business Banking Analyst Intern (Client: Banco Pichincha)** January 2026 - Present
NTT DATA *Quito*
 - Conduct root cause analysis and defect triage for enterprise banking applications by analyzing complex logs in Azure Kubernetes Service (AKS) using OpenLens.
 - Verify trace integrity and perform advanced database troubleshooting using SSMS and Toad Oracle via CyberArk for secure access.
 - Document incident resolutions and technical procedures in Confluence, ensuring strict adherence to ITIL v4 best practices.
 - Collaborate directly with cross-functional teams to identify system anomalies, track bugs, and improve overall operational stability.
- **Visual Programming Instructor** August 2024 - Present
Algorithmics *Quito*
 - Teach visual programming and video game design classes to children and teenagers aged 7 to 17 years.
 - Instruct technologies such as Scratch, Roblox Studio, and Python, adapting approaches for different ages.
 - Implement interactive educational materials to facilitate learning programming concepts.
- **Development Intern** June 2025 - August 2025
BIESS (Ecuadorian Social Security Institute Bank) *Quito*
 - Designed and implemented automated end-to-end testing with Python, Selenium, and Playwright, reducing deployment time by **40%**.
 - Integrated automated testing into CI/CD pipelines using Jenkins, improving error detection by **60%**.
 - Automated feature delivery report generation with Python, obtaining SonarQube metrics.
 - Reduced technical documentation and reporting time by **50%** through process automation.
- **Project Leader and Developer** August 2024 - December 2024
Radio Área Deportiva *Quito*
 - Led multidisciplinary team for planning and development of a web platform.
 - Implemented Scrum framework, reducing development time by **20%** and increasing online traffic by **35%**.
 - Developed responsive web architecture, improving user experience on mobile devices.

EDUCATION

- **Software Engineering (Final Semester)** September 2020 - Present (Est. 2026)
Escuela Politécnica Nacional *Quito*
 - Member of "Imagen Sistemas" group.
 - Focused on Software Quality, Secure Architecture, and Project Management.

CERTIFICATIONS

- **QA Innovation Lab Bootcamp** 2026
NTT DATA Online
- **Microsoft Certified: Azure Fundamentals (AZ-900)** 2026
Microsoft Online
- **GitHub Certification (GH-300)** 2026
GitHub / Microsoft Online
- **Microsoft Azure DevOps Engineer (AZ-400) Bootcamp** Nov 2024 - Feb 2025
Código Facilito Online
- **English Certification - Level B2 (Upper Intermediate)** Mar 2021 - Dec 2022
CEC EPN Quito

FEATURED PROJECTS

- **Denuncia Segura EC:** Engineered a secure reporting platform enabling citizens to anonymously report urban issues to the government, prioritizing whistleblower safety. Designed with a Microservices architecture (Java, Spring Boot) and an Angular frontend. Implemented Defense in Depth strategies including OAuth2, Magic Bytes Validator, and Role Guards. Deployed on Azure Container Apps with automated CI/CD via GitHub Actions, integrating SonarQube for continuous code quality inspection and Grafana/Loki for real-time logging and monitoring.
- **Educational AI Content Moderator Extension (Thesis Project):** Developing a custom browser extension integrated with an AI-driven backend to detect and filter potentially abusive content on online educational platforms. Focused on model accuracy validation, continuous testing, and ensuring ethical AI guidelines to maintain safe digital learning environments.
- **Facial Recognition System with Artificial Intelligence (2023):** Developed Siamese neural network for facial recognition achieving **95% accuracy**. Integrated system with Telegram and Alexa APIs for real-time access control. Stack: Python, TensorFlow.

TECHNICAL SKILLS

- **Testing & QA:** Test automation, API, End-to-End, Regression, Integration, and Performance Testing; tools: Postman, Pytest, Karate, Serenity BDD, Cucumber, Gherkin, Selenium, Playwright, Appium, K6; code quality with SonarQube.
- **Programming Languages:** Python, Java, JavaScript, C#, SQL.
- **Software Architecture & Frameworks:** Hexagonal Architecture, Microservices Architecture, Spring Boot, Angular, React.
- **DevOps & Cloud:** CI/CD and containerization with Git, GitHub Actions, Jenkins, Docker; cloud and orchestration with Azure DevOps, Azure Container Apps, Azure Kubernetes Service (AKS); monitoring with Grafana and OpenLens.
- **Agile Frameworks & Practices:** ISTQB, Scrum, Agile Development, TDD, BDD, ITIL v4.
- **Security:** Defense in Depth, OAuth2.

VOLUNTEER EXPERIENCE

- **Collaborator** September 2023 - September 2024
Grupo "Imagen Sistemas", Escuela Politécnica Nacional Quito
 - Organized and coordinated academic and scientific events for the faculty.
 - Coordinated soft skills workshops for university students.
 - Managed comprehensive planning, logistics, and execution of academic and professional events.
- **Digital Literacy Instructor** November 2023 - March 2024
Community Program for Senior Adults Quito
 - Taught basic computer skills and smartphone usage to adults over 60 years old.
 - Developed adapted educational materials to facilitate technological learning.
- **Animal Welfare Volunteer** March 2018 - December 2019
Fundación "Patitas y Sonrisas" Quito
 - Participated in rescue and rehabilitation programs for street dogs.
 - Collaborated in responsible adoption and sterilization campaigns.